

# **STA 130A: Mathematical Statistics: Brief Course**

## **Lecture 2: Set Theory and Probabilistic Models**

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# Announcements

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## Office hours:

- Instructor: Wed 2–3 PM (MSB 1143)
- TA: Thu 9–11 AM (MSB 1117)
- \* Please also use Piazza for course-related questions

## Important dates:

- Midterm 1: Mon, Oct 19 (in class)
- Midterm 2: Mon, Nov 9 (in class)
- Final: Mon, Dec 7, 3:30–5:30 PM
- \* *No make-up exams; see the syllabus for exam and accommodation policies*

**SDC accommodations:** If you need accommodations, please submit requests through the [Student Disability Center \(SDC\)](#) as early as possible

\* See the [course webpage](#) and [syllabus](#) for more details and additional information

# Announcements: Homework 1

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## Homework 1 is now posted

- Due: Thu, October 1, 11:59 PM PT
  - Late submissions will not be accepted for any reason and will receive 0 points
- Submission instructions:
  - You may typeset your solutions in  $\text{\LaTeX}$  (preferred), use a word processor, or handwrite and scan them; in all cases, make sure your work is legible and clearly organized
  - Upload a **single PDF file** to Canvas (*Assignments*  $\rightarrow$  *Homework 1*)
  - Name the file using the prefix of your UC Davis email ID and homework number (e.g., dgsong\_hw1.pdf)
  - Please make sure to include “STA 130A,” your name, and the last four digits of your student ID on the front page

\* See the [syllabus](#) for more details about homework policy and instructions

# Agenda

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- Set theory
- Probabilistic models, probability laws, and basic properties
- Models vs. reality

# Set theory: Notation and terminology

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A **set** is a collection of distinct objects, called the **elements** of the set

- $x \in S$ :  $x$  is an element of  $S$
- $x \notin S$ :  $x$  is not an element of  $S$

The **empty set** is a set having no elements, denoted by  $\emptyset$

Sets can be specified in various ways

- Roster notation (enumeration):  $S = \{x_1, x_2, \dots, x_n\}$ , or  $S = \{x_1, x_2, \dots\}$

e.g.,  $\{H, T\}$ ,  $\{1, 2, 3, 4, 5, 6\}$ ,  $\{2, 4, 6, 8, \dots\}$

- Set-builder notation (logical formula):  $S = \{x \mid x \text{ satisfies } Q\}$

e.g.,  $\{2k \mid k \text{ is a positive integer}\}$ ,  $\{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$

Sets may be finite, countably infinite, or uncountable

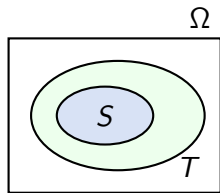
## Set theory: Inclusion relations

$S$  is a **subset** of  $T$  if every element of  $S$  is an element of  $T$ , i.e.,  $x \in S \implies x \in T$

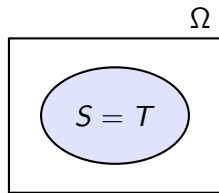
- We write  $S \subseteq T$  (or  $S \subset T$ )
- Equivalently,  $T$  is a **superset** of  $S$ , denoted by  $T \supseteq S$  (or  $T \supset S$ )

The two sets are **equal**, written  $S = T$ , if  $S \subseteq T$  and  $T \subseteq S$

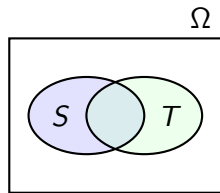
$S$  is a **proper (strict)** subset of  $T$ , denoted by  $S \subsetneq T$ , if  $S \subseteq T$  and  $S \neq T$



$S \subsetneq T$



$S = T$



Neither  $S \subseteq T$  nor  $T \subseteq S$

\* *Remark:* The inclusion relation  $\subseteq$  gives a partial order between sets

# Set theory: Set operations

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Often we can fix the **universal set (universe)**  $\Omega$ , which contains all objects of interest

- $S^c = \{x \in \Omega \mid x \notin S\}$  is the **complement** of  $S$  with respect to  $\Omega$

Given two sets  $S$  and  $T$ ,

- $S \cup T = \{x \mid x \in S \text{ or } x \in T\}$  is their **union**
- $S \cap T = \{x \mid x \in S \text{ and } x \in T\}$  is their **intersection**

Two sets  $S$  and  $T$  are **disjoint** if  $S \cap T = \emptyset$

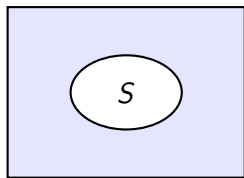
- A collection of sets is **pairwise disjoint** if no two distinct sets have a common element

A collection of sets  $\{S_1, S_2, \dots\}$  is a **partition** of  $S$  if

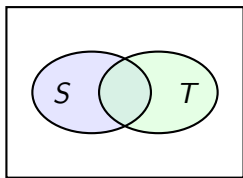
- $\bigcup_n S_n = S$  and
- the sets  $S_1, S_2, \dots$  are pairwise disjoint

## Illustration with Venn diagrams: Set operations

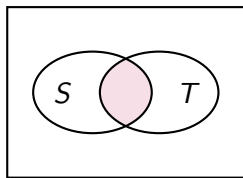
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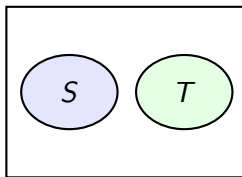
$S^c$



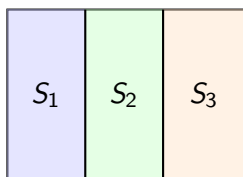
$S \cup T$



$S \cap T$



Disjoint



Partition of  $\Omega$

# Set theory: The algebra of sets

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Set operations have several properties, which are direct consequences of the definitions

- Commutativity:

$$S \cup T = T \cup S, \quad S \cap T = T \cap S$$

- Associativity:

$$S \cup (T \cup U) = (S \cup T) \cup U, \quad S \cap (T \cap U) = (S \cap T) \cap U$$

- Distributivity:

$$S \cap (T \cup U) = (S \cap T) \cup (S \cap U), \quad S \cup (T \cap U) = (S \cup T) \cap (S \cup U)$$

- ... and more:

$$(S^c)^c = S, \quad S \cap S^c = \emptyset, \quad S \cup S^c = \Omega, \quad S \cup \Omega = \Omega, \quad S \cap \Omega = S$$

# Set theory: De Morgan's laws

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The following properties are known as **De Morgan's laws**:

$$\left(\bigcup_n S_n\right)^c = \bigcap_n S_n^c, \quad \left(\bigcap_n S_n\right)^c = \bigcup_n S_n^c$$

Venn diagrams help visualize these identities for two or three sets

However, formally establishing these laws requires a *rigorous mathematical proof*

- To show  $(\bigcup_n S_n)^c \subseteq \bigcap_n S_n^c$ : *"Want: every  $x$  in  $(\bigcup_n S_n)^c$  also belongs to  $\bigcap_n S_n^c$ "*
  - Take  $x \in (\bigcup_n S_n)^c$
  - Then  $x \notin \bigcup_n S_n$ , which implies that for every  $n$ ,  $x \notin S_n$
  - Thus,  $x \in S_n^c$  for all  $n$ , and therefore,  $x \in \bigcap_n S_n^c$
- To show  $(\bigcup_n S_n)^c \supseteq \bigcap_n S_n^c$ : proceed similarly (*Exercise!*)

# Elements of a probabilistic model

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A **probabilistic model** is a mathematical description of an uncertain situation

- **Experiment**: an underlying process producing exactly one out of several possible outcomes

A probabilistic model consists of a sample space  $\Omega$  and a probability law  $P$

- **Sample space**  $\Omega$ : the set of all possible outcomes
  - Event<sup>1</sup>: a subset  $A \subseteq \Omega$ ; it occurs when the realized outcome  $\omega$  belongs to  $A$
- **Probability law**  $P$ : a map that assigns a number  $P(A)$  encoding our knowledge or belief about the collective likelihood of the event  $A$ , satisfying *certain axioms*

For a legitimate and useful probabilistic model,  $\Omega$  and  $P$  must satisfy some requirements

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<sup>1</sup>For uncountable sample spaces, events belong to a suitable collection of subsets; such measure-theoretic details are beyond the scope of this course.

# Choosing an appropriate sample space

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When choosing a sample space  $\Omega$ , keep three principles in mind:

- **Collectively exhaustive:**  $\Omega$  should include every outcome the experiment can produce
  - Choosing  $\Omega = \{1, 2, 3, 4, 5\}$  for a die roll will miss the outcome 6
- **Mutually exclusive:** the outcomes in  $\Omega$  should be distinguishable from one another
  - The sample space for a die roll cannot have both "1" and "1 or 4" simultaneously as possible outcomes
- **No unnecessary detail:** make  $\Omega$  fine enough to answer the questions of interest, but no finer
  - The landing location or maximum height is unnecessary if we only care about heads or tails

# Probability laws

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Once the sample space  $\Omega$  has been chosen, a probability law assigns to each event  $A$  a number  $P(A)$ , called the **probability** of  $A$ , subject to three axioms:

1. **Nonnegativity:**  $P(A) \geq 0$  for every event  $A$
2. **(Countable) Additivity:** if  $A$  and  $B$  are disjoint events, then

$$P(A \cup B) = P(A) + P(B)$$

More generally, if  $A_1, A_2, \dots$  are pairwise disjoint, then

$$P\left(\bigcup_n A_n\right) = \sum_n P(A_n)$$

3. **Normalization:**  $P(\Omega) = 1$

## Some properties of probability laws

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There are many natural properties of a probability law, derivable from the axioms

- $P(A^c) = 1 - P(A)$
- $P(\emptyset) = 0$
- If  $A \subseteq B$ , then  $P(A) \leq P(B)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A \cup B) \leq P(A) + P(B)$  (union bound)
- $P(B \setminus A) = P(B) - P(A \cap B)$   $B \setminus A := B \cap A^c$
- ...

It is often useful to visualize and verify these using a Venn diagram  
(*You may also want to prove them!*)

# Constructing a probability law: Example 1

## Example (Single coin toss)

**Experiment:** Tossing a coin once

**Possible outcomes:** Heads (H) and tails (T)

**Sample space:**  $\Omega = \{H, T\}$

**Events:**  $\{H, T\}, \{H\}, \{T\}, \emptyset$

If the coin is fair—meaning that we know or believe that heads and tails are “equally likely”—then we should assign equal probabilities to the two possible outcomes:  $P(\{H\}) = P(\{T\})$

The additivity and the normalization axioms together imply that

$$P(\{H, T\}) = P(\{H\}) + P(\{T\}) = 1$$

**Conclusion:** The resulting probability law is given by

$$P(\{H, T\}) = 1, \quad P(\{H\}) = \frac{1}{2}, \quad P(\{T\}) = \frac{1}{2}, \quad P(\emptyset) = 0$$

## Constructing a probability law: Example 2

### Example (Three coin tosses)

**Experiment:** Tossing a coin three times

**Possible outcomes:** length-3 sequences of H's and T's

**Sample space:**  $\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

**Events:** subsets of  $\Omega$  ( $2^8 = 256$  subsets)

We model the 8 sequences as equally likely (as for independent tosses of a fair coin), so each has probability  $1/8$

Consider an event  $A = \{\text{exactly 2 heads occur}\}$ : the resulting probability law gives

$$\begin{aligned} P(A) &= P(\{HHT, HTH, THH\}) \\ &= P(\{HHT\}) + P(\{HTH\}) + P(\{THH\}) \\ &= \frac{1}{8} + \frac{1}{8} + \frac{1}{8} \\ &= \frac{3}{8} \end{aligned}$$

## Discrete vs. continuous models

In a discrete model, the probabilities of singleton events determine the probability law:

$$P(A) = \sum_{a \in A} P(\{a\}).$$

In a continuous model, singleton probabilities typically do *not* determine the law

### Example (A wheel of fortune)

**Experiment:** Spin a fair wheel and record a number in  $[0, 1]$

**Possible outcomes:** the numbers in the interval  $[0, 1]$

**Sample space:**  $\Omega = [0, 1]$

**Events:** subsets of  $[0, 1]$

**Q:** What is the probability  $P(\{x\})$  for a single point  $x \in [0, 1]$ ?

- Under the natural uniform model on  $[0, 1]$ ,  $P(\{x\}) = 0$  for every  $x$
- Yet intervals can still have positive probability;  $P([a, b]) = b - a$  for  $0 \leq a \leq b \leq 1$

# Models vs. reality

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Using the framework of probability theory involves two distinct stages:

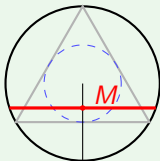
1. **Modeling:** Choose a sample space and a probability law to represent reality
  - The axioms ensure mathematical consistency; suitability depends on the situation
  - Different reasonable models may represent the same situation differently
  - Even a simplified “incorrect” model may be useful due to simplicity and tractability
2. **Analysis:** Once the model is fixed, compute probabilities and draw conclusions (*within the model*)
  - This stage is governed by the axioms of probability and ordinary logic
  - The main difficulty is often computation, not ambiguity

Some apparent paradoxes arise from an ambiguous or incomplete probabilistic model!

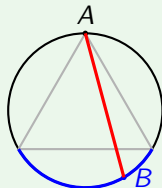
## Example: Bertrand's paradox

### Example (Bertrand's paradox)

**Question:** Given a circle and an inscribed equilateral triangle, what is the probability that the length of a randomly chosen chord of the circle is greater than the side of the triangle?



(a) Choose a random point uniformly along radius. A chord  $M$  is longer than the triangle side iff its midpoint lies within distance  $R/2$  of the center. Hence  $P = 1/2$ .



(b) Fix one endpoint  $A$ , then choose the other endpoint uniformly on the circumference. The favorable arc has length  $1/3$  of the circumference. Hence  $P = 1/3$ .

The phrase *“a randomly chosen chord”* does not specify a unique probabilistic model.

**Takeaway:** Different models of a “random chord” give different answers.  
*Probability theory itself is not inconsistent here.*

# Wrap-up

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## Set theory

- Set theory provides a formal language for probability theory; notation and basic operations
- Venn diagrams are useful for intuition, but mathematical proofs come from the definitions

## Probabilistic models

- A probabilistic model consists of a sample space  $\Omega$  and a probability law  $P$
- Choose  $\Omega$  to be exhaustive, mutually exclusive, and no finer than needed
- $P$  must satisfy the three axioms: nonnegativity, countable additivity, and normalization
- In discrete models, singleton probabilities determine  $P$ ; in continuous models, they may not

## Models vs. reality

- Modeling requires judgment: the same real situation can admit different reasonable models
- Once the model is fixed, probability calculations are governed by the axioms and logic

**Suggested reading:** [BT08, Ch. 1.1 & 1.2]

**Next lecture:** Conditional probability, Bayes' rule, and an introduction to independence.

# References

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Dimitri Bertsekas and John N Tsitsiklis.

*Introduction to probability*, volume 1.

Athena Scientific, 2nd edition, 2008.